Algebraic Topology and Algebra: Products, Wedge Sums, CW Complexes, Cosets, and Surface Gluings

N-20221229

§1.1 From quotient pictures to algebraic invariants

Several threads meet in one progression: the torus as $(\mathbb{R}/\mathbb{Z})^2$, the product $S^1 \times S^1$, disjoint unions and wedge sums, CW complexes, cosets, and polygonal gluing descriptions of surfaces. The important experiment is to attach a single two-cell to the same figure-eight one-skeleton in two different ways. The resulting torus and Klein bottle have identical cell counts but different fundamental groups and homology.[1]

A useful geometric realization of the product is

$$S^1 \times S^1 \cong T^2.$$

Think of a torus as a vertical circle carried around a horizontal circle. The path traced by that moving circle is the torus. This is not merely a picture: it is a parametrization.

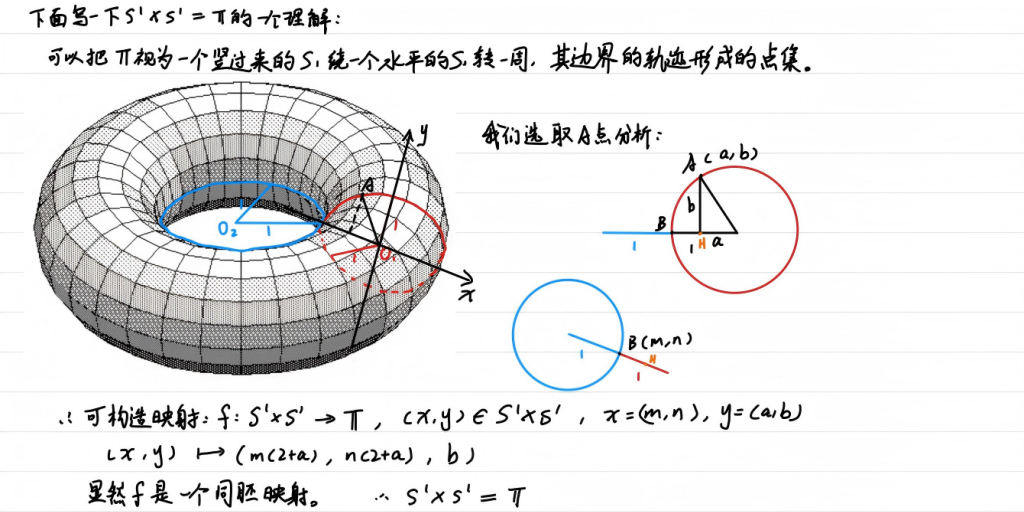


Figure 1.1: The two circle coordinates of the torus: one travels around its major circle, the other around its cross-section.

§1.2 $S^1 \times S^1$ as a parametrized torus

Write a point of the first circle as an angle θ, and a point of the second circle as an angle ϕ. For a torus with major radius R and minor radius r, a standard parametrization is

$$F(\theta, \phi) = ((R + r\cos\phi)\cos\theta, (R + r\cos\phi)\sin\theta, r\sin\phi).$$

[1] The CW construction, attaching words, cellular boundaries and projective-space examples follow A. Hatcher, *Algebraic Topology*, Chapters 0–2.

This realizes $S^1 \times S^1 \to T^2$. The first angle decides where the circular cross-section is placed around the central hole; the second decides where we are on that cross-section.

A point A may be decomposed into two circle parameters, but the invariant statement is cleaner: the torus is the space of ordered pairs of circle-points, realized geometrically by the map above.

§1.3 Disjoint union

For topological spaces X and Y, the disjoint union $X \amalg Y$ keeps the two spaces formally separate. A subset $U \subseteq X \amalg Y$ is open iff

$$U \cap X \text{ is open in } X, \qquad U \cap Y \text{ is open in } Y.$$

If X and Y are nonempty, then $X \amalg Y$ is disconnected. Indeed, X and Y are both open in the disjoint union topology, disjoint, nonempty, and their union is the whole space. This proof is more precise than the vague statement that the spaces are simply "side by side". The topology, not only the picture, gives the separation.

§1.4 Wedge sum

The wedge sum is obtained from a disjoint union by identifying chosen basepoints:

$$X \vee Y = (X \amalg Y)/(x_0 \sim y_0).$$

Thus disjoint union places spaces side by side, while wedge sum fuses them at one point. The prototype is $S^1 \vee S^1$, the figure-eight space.

Example 57.4.4 ($S^1 \vee S^1$ is a figure eight)
Let $X = S^1$ and $Y = S^1$, and let $x_0 \in X$ and $y_0 \in Y$ be any points. Then $X \vee Y$ is a "figure eight": it is two circles fused together at one point.

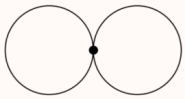


Figure 1.2: The wedge sum $S^1 \vee S^1$ as two circles fused at one point.

This construction matters because loops in the two circles remain visibly independent. Algebraically, the fundamental group of the figure eight is the free group on two generators. So the picture is not a toy: it is one of the first spaces where topology produces noncommutative algebra.

§1.5 CW complexes: building spaces by cells

A CW complex is built inductively. Start with a set of 0-cells. Attach 1-cells by gluing endpoints to the 0-skeleton. Attach 2-cells by gluing the boundary circle of a disk to the 1-skeleton. Continue dimension by dimension:

$$X^0 \subseteq X^1 \subseteq X^2 \subseteq \cdots.$$

A k-cell is a copy of D^k attached along S^{k-1} by an attaching map

$$\varphi_\alpha : S^{k-1}_\alpha \to X^{k-1}.$$

Informally: attach disks along their boundaries.

The disk D^2 illustrates that the same space may have different CW structures. One uses two 0-cells, two 1-cells, and one 2-cell. The other uses one 0-cell, one 1-cell whose two endpoints are glued to that point, and one 2-cell. Both build a disk, but the second is more economical.

Example 57.5.2 (D^2 with $2+2+1$ and $1+1+1$ cells)

(a) First, we start with X^0 having two points e_a^0 and e_b^0. Then, we join them with two 1-cells D^1 (green), call them e_c^1 and e_d^1. The endpoints of each 1-cell (the copy of S^0) get identified with distinct points of X^0; hence $X^1 \cong S^1$. Finally, we take a single 2-cell e^2 (yellow) and weld it in, with its boundary fitting into the copy of S^1 that we just drew. This gives the figure on the left.

(b) In fact, one can do this using just $1+1+1=3$ cells. Start with X^0 having a single point e^0. Then, use a single 1-cell e^1, fusing its two endpoints into the single point of X^0. Then, one can fit in a copy of S^1 as before, giving D^2 as on the right.

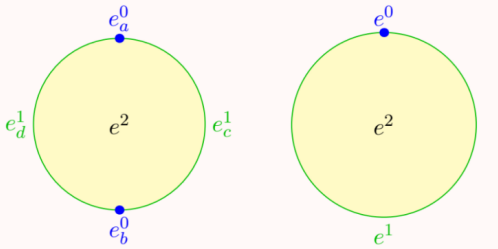


Figure 1.3: Two CW descriptions of D^2: one more literal, one more economical.

The lesson is that a CW decomposition is not unique. It is chosen because it makes later topology and algebra computable.

§1.6 The torus as a quotient of a square

The torus can be formed from a square by identifying opposite edges in the same direction. Walking off the right edge returns one at the corresponding point on the left edge; walking off the top returns one at the bottom. Algebraically,

$$T^2 \cong (\mathbb{R}/\mathbb{Z})^2 \cong S^1 \times S^1.$$

The quotient square viewpoint treats $[0,1] \times [0,1]$ as a fundamental domain for $(\mathbb{R}/\mathbb{Z})^2$.

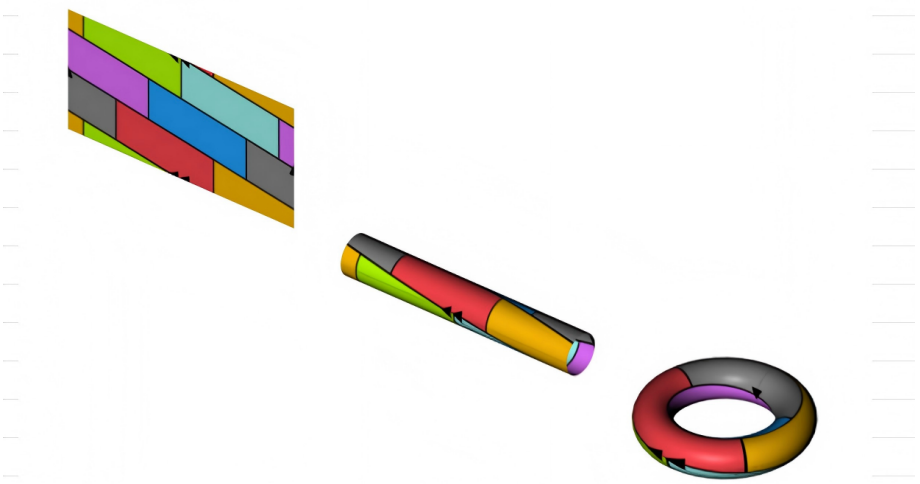


Figure 1.4: The quotient square/cylinder/torus picture: $(\mathbb{R}/\mathbb{Z})^2$ as a glued square.

As a CW complex, the torus has one 0-cell, two 1-cells a, b, and one 2-cell. The boundary of the 2-cell is attached along the word

$$aba^{-1}b^{-1}.$$

This gives the familiar presentation

$$\pi_1(T^2) = \langle a, b \mid aba^{-1}b^{-1} = 1 \rangle \cong \mathbb{Z}^2.$$

The commutativity of $\mathbb{Z}^2$ is encoded by the boundary word of the square.

§1.7 Cosets as algebraic quotients

Cosets are the algebraic analogue of quotienting. If H is a subgroup of G, right congruence modulo H is

$$a \equiv_r b \pmod{H} \iff ab^{-1} \in H.$$

It is an equivalence relation. Reflexivity uses $aa^{-1} = e \in H$; symmetry uses $(ab^{-1})^{-1} = ba^{-1}$; transitivity uses

$$ac^{-1} = (ab^{-1})(bc^{-1}).$$

The equivalence class of a is the right coset

$$Ha = \{ha : h \in H\}.$$

The map $H \to Ha$, $h \mapsto ha$, is a bijection, so $|Ha| = |H|$. The group G is partitioned into cosets, and two cosets are either equal or disjoint. This is exactly the algebraic version of cutting a space into equivalence classes.

The quotient $(\mathbb{R}/\mathbb{Z})$ illustrates the same principle in a geometric setting. Both are quotient constructions: identify things differing by a subgroup.

§1.8 The Klein bottle

The Klein bottle is built like the torus, except one pair of edges is identified in the opposite direction. If one first glues a pair of edges, one gets a cylinder. But gluing

the resulting boundary circles in opposite directions cannot be embedded in ordinary three-dimensional space without self-intersection. The usual picture in $\mathbb{R}^3$ is therefore an immersion, not a faithful embedding.

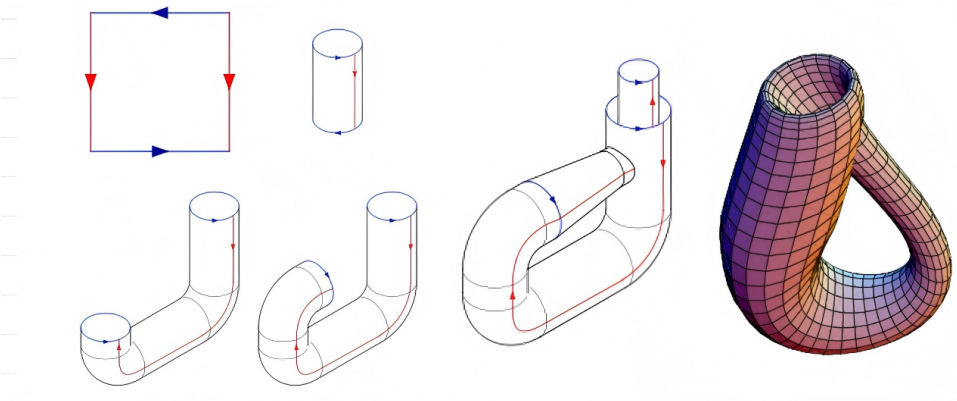


Figure 1.5: The Klein bottle: one edge identification is reversed, so the usual 3D picture self-intersects.

As a CW complex, the Klein bottle again has one 0-cell, two 1-cells, and one 2-cell, but the attaching word changes. A common presentation is

$$\pi_1(K) = \langle a, b \mid aba^{-1} = b^{-1} \rangle.$$

The relation says that going around one direction reverses the other. Thus the torus and Klein bottle have the same cell counts but different attaching words, and this difference controls orientability and fundamental group.

§1.9 Polygon schemata for surfaces

Labelled polygons give concrete gluing schemata for surfaces. A polygon with labelled, oriented boundary edges can be glued according to those labels. The resulting quotient is a surface. For example, the torus corresponds to

$$aba^{-1}b^{-1},$$

while an orientable surface of genus g has a boundary word of the form

$$a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}.$$

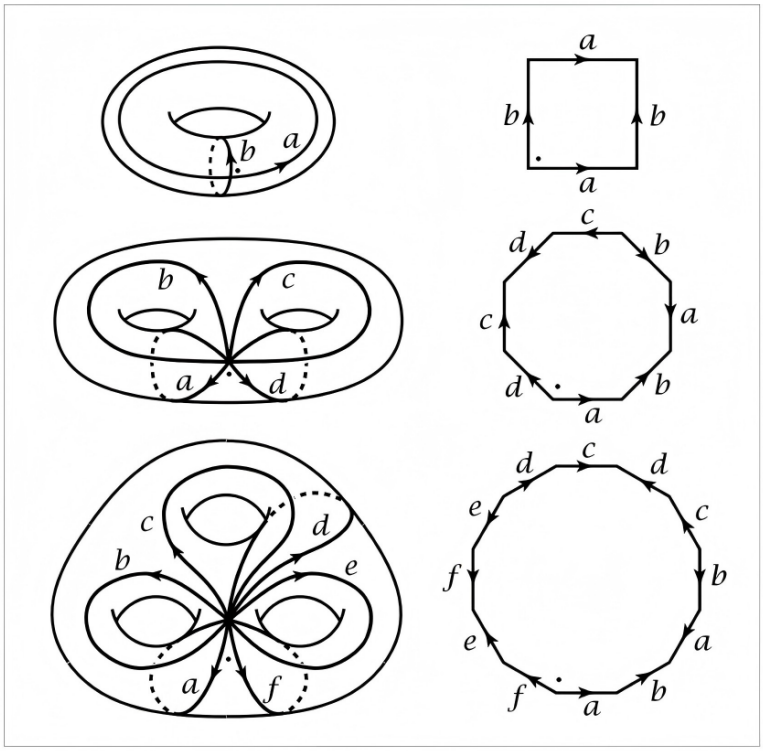


Figure 1.6: Polygon schemata and surface gluings: topology encoded by labelled boundary words.

The picture compresses several ideas at once: cut a surface into a polygon, label the boundary, glue equal labels, and read those labels as algebraic generators and relations.

§1.10 Projective spaces as quotient spaces

Real and complex projective spaces give two further quotients with strikingly different cellular structures. Real projective space is

$$\mathbb{RP}^n = S^n/(x \sim -x),$$

the space of lines through the origin in $\mathbb{R}^{n+1}$. Complex projective space is the space of complex lines through the origin in $\mathbb{C}^{n+1}$, equivalently

$$\mathbb{CP}^n = S^{2n+1}/S^1.$$

A useful CW fact is that $\mathbb{RP}^n$ has one cell in each dimension $0, 1, \ldots, n$, while $\mathbb{CP}^n$ has one cell in each even real dimension $0, 2, 4, \ldots, 2n$.

§1.11 An attaching word has two algebraic interpretations

Begin with the figure eight

$$X^1 = S^1 \vee S^1.$$

Its fundamental group is the free group $F(a, b)$ on the loops a, b. A cellular map $S^1 \to X^1$ can be described, up to homotopy, by a word $w(a, b)$: as one goes around the boundary circle, the map traverses the two loops with the orientations specified by the letters.

Attach a two-cell along this map, forming $X = X^1 \cup_w D^2$. Van Kampen's theorem says that the attaching disk makes the loop w null-homotopic:

$$\boxed{\pi_1(X) \cong F(a,b)/\langle\langle w \rangle\rangle,}$$

where the denominator is the normal closure of the word. The normal closure matters: if a loop becomes null-homotopic, so do all its conjugates.

Now compute the same attaching map in cellular homology. Since X has one cell in dimension zero, two in dimension one, and one in dimension two, its cellular chain complex is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\partial_2} \mathbb{Z}^2 \xrightarrow{\partial_1 = 0} \mathbb{Z} \longrightarrow 0.$$

The boundary of the two-cell is obtained by *abelianizing the word*: if w has exponent sums r in a and s in b, then

$$\boxed{\partial_2(1) = (r, s).}$$

The group computation remembers the order of the letters; the homology computation remembers only their total signed counts. This difference is exactly why the same attaching diagram contains more information than the resulting chain complex.

§1.12 Torus versus Klein Bottle

For the torus, the boundary word is

$$w_T = aba^{-1}b^{-1}.$$

Its exponent sums are both zero, hence $\partial_2 = 0$. It follows that

$$\begin{aligned} H_0(T^2;\mathbb{Z}) &\cong \mathbb{Z}, \\ H_1(T^2;\mathbb{Z}) &\cong \mathbb{Z}^2, \\ H_2(T^2;\mathbb{Z}) &\cong \mathbb{Z}. \end{aligned}$$

In the fundamental group, however, the attaching relation is $aba^{-1}b^{-1} = 1$, or $ab = ba$, so

$$\pi_1(T^2) = \langle a, b \mid [a,b] = 1 \rangle \cong \mathbb{Z}^2.$$

For the Klein bottle, take the standard word

$$w_K = aba^{-1}b.$$

It has exponent sum zero in a and two in b. Therefore

$$\partial_2(1) = (0,2), \qquad H_1(K;\mathbb{Z}) = \frac{\mathbb{Z}^2}{\langle (0,2) \rangle} \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z},$$

whereas

$$H_2(K;\mathbb{Z}) = \ker \partial_2 = 0.$$

The corresponding fundamental-group presentation is

$$\pi_1(K) = \langle a, b \mid aba^{-1} = b^{-1} \rangle.$$

It records a conjugation that reverses one generator. That reversal is the algebraic shadow of the twist in the edge identification and of the failure of orientability.

Both spaces have Euler characteristic

$$\chi = 1 - 2 + 1 = 0.$$

Thus χ is incapable of distinguishing the two gluing patterns. The chain complex can distinguish them because it remembers ∂_2, and the fundamental group distinguishes them even more sharply because it remembers the noncommutative attaching word.

There is a second instructive comparison. The figure eight itself has

$$H_1(S^1 \vee S^1; \mathbb{Z}) = \mathbb{Z}^2$$

just like the torus. The difference is the extra two-cell: its attaching commutator is invisible after abelianization, so H_1 does not change, but $H_2 = \mathbb{Z}$ appears and π_1 becomes abelian. This shows exactly which information each invariant forgets.

§1.13 Projective Spaces in All Dimensions

Real and complex projective spaces are quotient spaces:

$$\mathbb{RP}^n = S^n/(x \sim -x), \qquad \mathbb{CP}^n = S^{2n+1}/S^1.$$

Their natural CW decompositions are rather different.

The space $\mathbb{RP}^n$ has one cell e^k in every dimension $0 \le k \le n$. Its k-skeleton is $\mathbb{RP}^k$. The attaching map of e^k is the double covering $S^{k-1} \to \mathbb{RP}^{k-1}$. When the cellular attaching map is projected onto the quotient of the $(k-1)$-skeleton by its $(k-2)$-skeleton, the resulting degree is

$$\boxed{\partial_k = 1 + (-1)^k.}$$

One way to see the sign is to compare the two hemispheres of S^{k-1}: the second contributes the degree of the antipodal map, which is $(-1)^k$. Thus the cellular boundaries alternate between multiplication by 2 and the zero map:

$$\partial_{2j} = 2, \qquad \partial_{2j+1} = 0.$$

For $\mathbb{RP}^2$, the chain complex is therefore

$$0 \longrightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0,$$

and

$$\boxed{H_1(\mathbb{RP}^2; \mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}, \qquad H_2(\mathbb{RP}^2; \mathbb{Z}) = 0.}$$

The 2-torsion is the homological residue of the antipodal identification.

Complex projective space has instead one cell in every even real dimension $0, 2, \dots, 2n$, and no odd-dimensional cells. Every cellular differential therefore vanishes for dimension reasons. Hence

$$\boxed{H_{2j}(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z} \quad (0 \le j \le n),} \qquad H_{2j+1}(\mathbb{CP}^n; \mathbb{Z}) = 0.$$

The difference in cell dimensions explains the difference in homology with almost no additional computation.

§1.14 Orientability is encoded by the top boundary

The chain computations give a uniform way to detect orientability for connected closed surfaces. A coherent orientation on all top-dimensional cells produces a nonzero fundamental cycle, which generates $H_2(M;\mathbb{Z}) \cong \mathbb{Z}$. If going around a loop reverses the local orientation, such a global cycle cannot exist, and $H_2(M;\mathbb{Z}) = 0$.

The torus has a top cycle because the opposite oriented edge contributions cancel. The Klein bottle does not: its top two-cell has boundary $2b$, so it cannot be a cycle over $\mathbb{Z}$. Nevertheless, with coefficients in $\mathbb{F}_2$, that boundary becomes zero:

$$2b = 0 \quad \text{in } \mathbb{F}_2.$$

Thus

$$H_2(K;\mathbb{F}_2) \cong \mathbb{F}_2,$$

as for every connected closed surface. Reduction modulo two forgets the distinction between positive and negative orientation.

The labelled polygon has led to a chain of increasingly sensitive invariants. The cell counts give Euler characteristic, exponent sums give cellular boundary maps, and the full boundary word gives the fundamental-group presentation. The arithmetic of 2-torsion and the sign of an attaching map retain information that an unlabelled drawing or an Euler-characteristic calculation cannot.